Arista Access Node HEB-ACC-11 Optical Transport Failure Standard Operating Procedure

Node: HEB-ACC-11 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_7726

Description

The failure of the optical transport on the Arista Access Node HEB-ACC-11 at the Hebbal Edge site is a complex phenomenon that arises from the intricate interplay of hardware and protocol mechanisms. The Arista 7150 series switches, employed in this scenario, utilize SFP+/XFP-based optical interfaces for high-speed data transmission. However, these interfaces are susceptible to degradation due to factors such as fiber cuts, Rx/Tx power loss, or SFP/XFP module failures. The alarm code ALARM_7726, indicative of a minor optical transport failure, is triggered when the switch detects a significant deviation in the received optical signal power or a failure in the SFP/XFP module. The underlying protocol mechanics, governed by the IEEE 802.3z standard, dictate the operation of the Gigabit Ethernet interfaces. In the event of an optical transport failure, the switch will automatically switch to a backup interface, if available, to ensure uninterrupted data transmission. Nevertheless, prompt intervention is essential to diagnose and rectify the issue to prevent potential network-wide disruptions. This SOP outlines the step-by-step procedure for diagnosing and resolving the optical transport failure on the Arista Access Node HEB-ACC-11.

Pre-requisite Checks

- Verify that the Hebbal Edge site is accessible and that remote hands are dispatched to facilitate on-site intervention.

- Check the status of redundant links to ensure that they are operational and can assume traffic in the event of a primary link failure.

- Gather all relevant documentation, including the switch configuration and inventory records of the SFP/XFP modules.

Potential Root Causes

Cause	Probability
SFP/XFP module failure	High
Fiber cut or damage	Medium
Rx/Tx power loss due to optical signal degradation	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces status, Expected Output: Interface Status Description, Description: This command will display the status of all interfaces on the switch, including the optical interfaces. The output will indicate if the interface is up or down, and provide additional details such as the interface speed and duplex mode.
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```
Command: show interface H1/1/0 detail, Expected Output: Interface H1/1/0 is down (Admin down), Description: This command will provide detailed information about the specific interface H1/1/0, including its administrative and operational status. The output will indicate if the interface is down due to a hardware or software issue.

Command: show log | include ALARM_7726, Expected Output: ALARM_7726: Optical transport failure on interface H1/1/0, Description: This command will search the system log for any entries related to the alarm code ALARM_7726. The output will indicate the specific interface that is experiencing the optical transport failure.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: reboot, Description: Reboot the switch to ensure that any changes made to the configuration or hardware take effect. Verify that the new SFP/XFP module is properly seated and secured in the interface slot.

Step 2: Step: Verify the fiber connections, Command: show interfaces status, Description: Re-check the status of the optical interfaces to ensure that they are operational. Verify that the fiber connections are secure and not damaged.

Step 3: Step: Adjust the Rx/Tx power levels, Command: interface H1/1/0, Description: Adjust the receive and transmit power levels on the affected interface to ensure that they are within the recommended range. This may require reconfiguring the interface settings or replacing the SFP/XFP module.

Post-Check Verification

```
Command: show interfaces status, Expected Output: Interface Status Description, Description: Verify that the optical interfaces are operational and that the issue has been resolved.

Command: show log | include ALARM_7726, Expected Output: , Description: Verify that the alarm code ALARM_7726 is no longer present in the system log.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: reboot, Description: Reboot the switch and restore the original SFP/XFP module to its original position in the interface slot.

- Step: Revert the Rx/Tx power levels, Command: interface H1/1/0, Description: Revert the receive and transmit power levels on the affected interface to their original settings.